

The dark secret

A CVS stores files in a central location which is accessible to all the users of the files. A copy of the file needs to be 'checked out' by a developer, who then works on it and will 'commit' the changes back into the central repository when he or she is finished. Files which are being moved between the developer's location and the repository are scanned for changes before updating to prevent the work of one programmer from being overwritten by another, which would be a major disaster.

Versioning on the Web

In large-scale projects which involve programmers from different locations all over the world, like open source projects, things are a little more complicated.

With developers working at odd hours and across different time zones, giving a developer exclusive rights to a file would prevent any other developer from working on that file. This would quickly frustrate developers who would find that they are unable to work on a file when they actually find it convenient to work.

KDE, and GNOME.

CVS tackles this problem with its unreserved 'checkout' model which means that a single developer does not get exclusive rights to a file. Other developers can also refer to a copy, make changes to it, and then check it back in.

To eliminate the possibility of loss of work due to overwriting of versions by work done by another developer, CVS detects when multiple developers make changes to the same file. The system then automatically merges those changes and updates them as long as the

CVS allows developers to retrieve old versions of code and track the bugs that might have crept in

This system ensures that a history of each file is retained—this allows you to track who made what change and why, so that at any given point of time, you can restore the file to one of its previous stages.

In order to save on the disk space that would be occupied if each version were to be saved as a separate file, CVS stores all the versions within a single file. This approach is very intelligent because it works by recording only the differences between the various versions.

Critical to Teamwork

Version control is imperative for software development teams. While individual developers want version control to ensure that they do not lose work, every member within a development team must have access to a data central server.

When all members of the team are working from a single office, this is pretty straightforward: you just need to set up the CVS server on a machine on the local network. A developer can 'check out' a file from the server to work on.

When he has completed the modifications that he needs to make, he just 'checks in' the modified file into the repository.

While a particular file has been 'checked out,' the system does not allow any other developer to 'check out' and modify the same file.

However, CVS's built-in client-server access model allows any developer who can connect to the Internet to access files on a CVS server anywhere on the globe. This is why CVS is popular with several open-source projects like Mozilla www.mozilla.org/cvs.html the GIMP, <http://cvs.xemacs.org/>, XEmacs,

changes have not been made to the same lines of code. As an additional safety measure, just in case the system is unable to safely resolve the changes that have been made to the code, the developer or the designated administrator for that set of files is asked to merge them manually.

A CVS for everyone

The epicentre of the Open Source movement is SourceForge.net which offers the single largest collection of applications and source codes on the Internet. SourceForge provides Open Source developers with free services like project hosting, bug and issue tracking, version control, and project archival.

SourceForge.net offers you access to a number of project development resources. All you need to do is create a new account. Amongst these is access to your own personal CVS repository, the contents of which can be accessed through the use of a Web-based interface.

The popularity of SourceForge, especially with Open Source developers, is really highlighted by the prominent projects that have found their home there. These include The Freenet Project, phpGroupWare, Python, Gaim and kicp to name a few.

Is version control for you?

While CVS is designed to address the needs of developers, it is versatile enough to be useful to others too. Writers, for instance, could use it as a means to back-up their work. If your PC is connected to the Internet a CVS is very useful when you are collaborating with different writers or need access when you are travelling far from home.

System administrators would also find it a boon because they could use it to keep track of critical configuration files. Changes to the files can be maintained in the CVS tree so that they always have the option to roll back to an older working file in case the changes that were made did not work exactly as planned. This is especially important to those system administrators who have to manage large server farms. They can maintain a CVS tree of the configurations for each server—this would mean that whenever a new server is added, they could quickly locate the **config** file for a similar machine and get it up and running in no time. 

KARAN MANRAL